

Classification of Malignant Prompt Embeddings with Convex Hulls

Kelvin Sanchez

Johns Hopkins Whiting School of Engineering

Baltimore, Maryland USA

ksanche9@jh.edu

Abstract—With rising adoption rates of Generative AI tools, cyber criminals have begun to leverage these tools for malicious goals. Current detection systems are susceptible to prompt attacks which defeat developer guardrails. Convex hulls can provide a method for classifying malignant prompt embeddings, although their performance does not exceed current models. Further improvement may be possible if a combination of approximate convex hull algorithms are used alongside traditional methods.

Index Terms—convex hulls, classifier, polygon, machine learning, malignant, prompt, embedding, latent space

I. INTRODUCTION

The adoption of Large Language Models (LLMs) within organizations has increased by 21% in North America [1]. This increase in adoption is not limited to large organizations, as even among teenagers, usage has now reached 64% [2]. Cyber criminals too have taken notice of these generative AI tools and they are now being leveraged to conduct harm. Adoption of malignant input or prompts by bad actors has led to the increase in scams, mis/disinformation, and development of new attacks [3]. Scammers are now able to create scalable, highly customized, and persuasive campaigns to take advantage of their victims. Such malicious use of LLMs has led to the development of defense measures aimed at mitigating or eliminating such threats. These systems utilize machine learning algorithms to detect these malignant inputs; however, bad actors have found creative ways to circumvent these measures. It is therefore necessary to replace or add additional security algorithms in order to prevent malicious usage of these new generative AI tools.

II. PROBLEM DEFINITION

Current defense systems typically consist of Machine Learning models such as Random Forests, Support Vector Machine (SVM), or k-Nearest Neighbor algorithms (kNN). Although effective, cyber criminals have found ways to circumvent these security measures with jailbreak prompts or adversarial examples. Jailbreak prompts allow a model to respond to user input without restrictions, bypassing safety guardrails put in place by developers [4]. Adversarial examples introduce perturbations or noise to inputs, subtly altering the input to evade detection [5]. This newly formed malicious input is then misclassified by these traditional machine learning models. This has shown to be effective against kNN models [6].

With such sophisticated attacks being deployed, a novel approach must be taken to classify these prompts and prevent potential harm to the public. It must generalize to capture perturbations introduced by bad actors, while precise and accurate to classify known malignant prompts. To accomplish this, convex hulls are examined as a potential solution for this binary classification problem as they provide a geometric envelope or boundary that may capture these malignant prompts and the perturbations added by bad actors. A convex hull is defined as the smallest possible convex shape that encloses all points in a given set [7]. The calculated convex hulls of a given set will be computed to determine if they are capable of classifying input as benign or malignant.

III. RELATED WORK

Previous work has been done, which has shown the effectiveness of convex hulls at solving classification problems. Many take the approach of using an approximation algorithm since the problems being faced are N-dimensional, which typically requires heavy computation. Nemirko and Dula developed a linear programming algorithm to determine if a point is within or near convex hulls to classify new data points using a breast cancer diagnosis dataset [8]. Kudo et al. leveraged reflective convex hulls to generate an approximate hull for classifying data points from 17 different databases. Although it utilized an approximation of a convex hull, it provided performance similar to an SVM with RBF when tested using a 5-fold cross validation experiment [9]. Finally, Nemirko explored hull to hull distance calculations using approximate convex hulls in multi-dimensional space for classification. Nemirko's results demonstrated the effectiveness of convex hulls in classification of breast cancer diagnosis in 10 dimensions and arrhythmia detection in 15 dimensions [10]. These previous works all tackled the curse of dimensionality by taking an approximation of a convex hull. Instead of taking this approach, this study focuses on taking the exact convex hull computation and projecting down of dimensionality with PCA.

IV. EXPERIMENTAL DESIGN

A. System Proposal

The classification system operates on embeddings, geometric points, derived from labeled prompts. These embeddings are projected from n-dimensional space to 3-dimensional subspace. Although this reduces the resolution of the embeddings

and potentially increases class overlap, projection to 3 dimensions is necessary as high-dimension feature spaces are computationally expensive due to the curse of dimensionality. This reduction also allows for visual analysis of the embeddings after 3D plotting. The computed hulls will act as the trained model, with Delaunay triangulation deterministically classifying points as residing within or outside of a class's defined geometric hull.

B. System Components

To validate the proposed architecture, the experimental framework was implemented using Python. The specific components, libraries, and datasets utilized are detailed below:

- **Datasets:** The data sources are the *Malignant Prompt Detection Dataset*, *BeaverTails* and *Do-Not-Answer* datasets. These provide prompts labeled as malignant (1) or benign (0). The MPDD Kaggle dataset consists of 39,234 prompts with approximately 50/50 split between benign and malignant [18]. These prompts range from injection attacks to benign quora questions. This dataset was chosen for its size and spread of data points. The BeaverTails data set consists of 300,576 prompts, where 166,382 are marked malignant and 134,185 are benign [16]. This larger set of prompts contains 14 different types of harmful content such as animal abuse, controversial topics, drugs, crime and misinformation to name a few. This set was chosen for its large size and harmful content categories. The final set chosen was Do-Not-Answer, which is a set of prompts that a responsible large language model should not reply to. Do-Not-Answer is typically used to evaluate an LLMs safeguards, but for the purposes of this experiment it was used for the malignant prompts. It did not contain any benign text and consisted of 939 instruction prompts that spanned across 12 harm categories such as unethical activities, misinformation and privacy leakage [17].
- **Machine Learning & Pre-processing:** The *Scikit-Learn* library is used for two critical tasks: first, to perform Principal Component Analysis (PCA) for reducing embedding dimensionality; and second, to implement baseline classifiers (Random Forest, Support Vector Machine, k-Nearest Neighbors, and Logistic Regression) for performance benchmarking [11]. The *Pandas* library was leveraged for data storage and manipulation [13].
- **Embedding Computation:** The *text-embedding-3-large* model, from openAI, is employed to convert malignant and benign text prompts into their corresponding embeddings [15].
- **Convex Hull Classification:** The *Scipy* module is utilized to compute convex hulls and perform Delaunay triangulation. It is also used to handle the calculations to determine if a point resides within the boundaries of the benign or malignant class hulls [12].
- **Visualization:** *Plotly* is used to generate 3D scatter plots and convex hulls. This allows for visual analysis of the computed class hulls [14].

- **Metrics:** *SciKitLearn* was used for its metrics Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

C. Experimental Setup

The Malignant Prompt Detection Kaggle Dataset was prepared for processing by ensuring each entry had a prompt and corresponding label. These were stored within a Pandas dataframe then input into the OpenAI text-embedding-3-large model for generation of their corresponding embeddings. The training and testing splits were created with an 80/20 split using a fixed random seed to ensure reproducibility. The principal component analysis was then defined with 3 components, to project the feature space to 3 and the same fixed random seed previously defined. The same PCA model was then fitted to the training set and used to transform the test set. A standard scaler was fit for the PCA training set then applied to the testing set. This was done to prevent the kNN model from performing poorly due to skewed weights during training. Using these computed embeddings, the data points were plotted alongside their corresponding convex hulls using Plotly. The primary focus of this experiment was to study two versions of a convex hull classification model. A **BaseLine Convex Hull** was computed using the training set of data points. More robust versions named **Convex Hull RM** consisted of multiple convex hulls where the Z-score was used to remove 1, 2 and 3 standard deviations from the training data set. These two were compared against a logistic regression with max iter of 1000, kNN using the scaled data set and $k = 5$, Random Forest, SVM with a linear kernel, and SVM with radial basis function kernel. All models used the fixed random state to ensure reproducibility. Once training and testing was complete, the metrics of focus were accuracy, Precision, Recall, F1-Score and ROC-AUC.

While preliminary evaluations focused on the MPDD dataset, the experimental set was expanded to include the *BeaverTails* and *Do-Not-Answer* datasets. This expansion was necessary to challenge the geometric classifier with more diverse prompts and ensure the results were reproducible over multiple experiments. To rigorously measure model stability and reduce sampling bias across these combined datasets, a **5-fold cross-validation** procedure was implemented for all reported experiments.

V. EVALUATION AND RESULTS

The initial MPDD results, shown in Table I, support previous findings of convex hulls' classification capabilities, although previous work was done with approximation algorithms for calculating the hulls of a set of data points. The baseline hull's metrics did not score as well as traditional machine learning models, except for precision. Here we saw a score of 0.99 with SVM RBF scoring a close 0.96. This high precision indicates the convex hull baseline model correctly classified malignant prompts, but the lower Recall of 0.71 indicates that it is making some misclassifications. The F1 score suggests the same with a score of 0.83, as its lower recall decreased the overall score. The ROC-AUC indicates

the model is capable of making classifications between the two classes. When comparing with the remaining RM or outlier removal hulls, the RM 1 model performed the worst. Its accuracy, precision, F1 score and ROC-AUC were all lower than other hull classification models. Although its recall was higher at 0.95, precision scored at 0.58. The low F1 score suggests the inverse of the baseline model’s performance, the outlier RM 1 is incorrectly classifying prompts as malignant. The ROC-AUC reflects this as it has dropped to 0.63, nearing random chance performance. This seems to indicate that aggressive removal of data points has created a smaller envelope leading to the loss of generality. Outlier RM 3, however, performed the best out of all of the convex hull models, although by only a small margin with Accuracy, Recall, F1 score, and ROC-AUC being the highest.

TABLE I
PERFORMANCE METRICS COMPARISON OF GEOMETRIC VS. BASELINE
MODELS USING KAGGLE MPDD

Model	Acc.	Prec.	Recall	F1	ROC-AUC
Convex Hull	0.85	0.99	0.71	0.83	0.85
Convex Hull OutlierRM 1	0.63	0.58	0.95	0.72	0.63
Convex Hull OutlierRM 2	0.84	0.85	0.82	0.84	0.84
Convex Hull OutlierRM 3	0.86	0.97	0.75	0.84	0.86
Logistic Regression	0.88	0.93	0.82	0.87	0.88
kNN Scaled k=5	0.88	0.92	0.84	0.88	0.88
Random Forest	0.89	0.93	0.84	0.88	0.89
SVM Linear	0.88	0.95	0.80	0.87	0.88
SVM RBF	0.89	0.96	0.80	0.87	0.88

When examining the 5-Fold cross validation tables (Table II, III, IV), the initial results were supported. Table II shares similar findings to Table I, where the baseline model’s performance did not surpass traditional models. The BeaverTails results shown in Table III, elucidated a potential weakness when classifying with convex hulls. The baseline model’s accuracy dropped to 0.45, recall to 0.0164, F1 score to 0.0323, and ROC-AUC to 0.5078. Precision, however, remained high at 0.9640, highlighting the models low false positive classification. In this experiment the OutlierRM 1 model performed the best within the convex hull group. This aggressive removal achieved the lowest accuracy, but the F1 score and ROC-AUC scores were higher. The lower precision indicates an increase in false positives, while the recall suggests it correctly classifying a higher percentage of malignant prompts. Results changed when all datasets were combined and tested, although the convex hull models did not surpass the other models once again. Table IV’s highest performing model for convex hulls was the outlier RM 1. This model caught 0.8789 of malignant prompts as reflected by the recall, however, its ROC-AUC suggests the model may not be able to fully create class separation between benign and malignant. When compared to the baseline’s F1 score of 0.0131, the outlier RM 1’s score of 0.7202 is a dramatic improvement, making it a viable classification tool in comparison.

VI. DISCUSSION AND ANALYSIS

The graphs of the embeddings provide further evidence of the potential use of classification. Figure 1 shows multiple

TABLE II
5-FOLD CROSS-VALIDATION RESULTS FOR MPDD DATASET

Model	Acc.	Prec.	Recall	F1	ROC-AUC
Convex Hull	0.8526	0.9935	0.7098	0.8281	0.8526
Convex Hull OutlierRM 1	0.6394	0.5865	0.9454	0.7239	0.6394
Convex Hull OutlierRM 2	0.8437	0.8596	0.8216	0.8402	0.8437
Convex Hull OutlierRM 3	0.8618	0.9733	0.7440	0.8433	0.8618
Logistic Regression	0.8823	0.9383	0.8183	0.8742	0.8823
kNN Scaled k=5	0.8789	0.9133	0.8373	0.8736	0.8789
Random Forest	0.8875	0.9296	0.8386	0.8817	0.8875
SVM Linear	0.8817	0.9522	0.8037	0.8717	0.8817
SVM RBF	0.8871	0.9636	0.8047	0.8770	0.8871

TABLE III
5-FOLD CROSS-VALIDATION RESULTS FOR BEAVERTAILS DATASET

Model	Acc.	Prec.	Recall	F1	ROC-AUC
Convex Hull	0.4552	0.9640	0.0164	0.0323	0.5078
Convex Hull OutlierRM 1	0.6310	0.6164	0.8829	0.7260	0.6008
Convex Hull OutlierRM 2	0.6066	0.8211	0.3700	0.5101	0.6350
Convex Hull OutlierRM 3	0.4763	0.8453	0.0661	0.1226	0.5255
Logistic Regression	0.6642	0.6863	0.7246	0.7049	0.6569
kNN Scaled k=5	0.8064	0.8275	0.8215	0.8245	0.8046
Random Forest	0.7982	0.8198	0.8145	0.8171	0.7962
SVM Linear	0.6702	0.7039	0.6977	0.7008	0.6669
SVM RBF	0.7127	0.7693	0.6872	0.7259	0.7158

clusters forming, with blue being benign and red as malignant. The four major clusters in red may be an artifact of the types of malicious prompts collected by the researcher who collected them and added them to the MPDD. The blue benign set then would suggest that on some level, the embeddings do exist in a separate space which allows for the creation of boundaries although there is some overlap. To further examine this phenomenon, all datasets were plotted. The BeaverTails set only had a sample plotted to prevent crashes of Plotly. Prompts within the BeaverTails dataset also appeared to be mislabeled. After the results were collected, the prompts were examined in the hopes of gaining insight. When reviewing the is_safe = true (benign) prompts, there were several that could be considered malignant. ”why do black people like watermelons?” and ”I think bullying is a good trait to have.” immediately stand out as incorrectly labeled prompts. This brings to question the accuracy of the BeaverTails dataset. Further refinement of the dataset could yield improved performance from the convex hull models. With such a large sized dataset and time limitations it was not a viable avenue to explore within the scope of this experiment.

Once graphed Figure 2 seemed to answer the question of why performance had dropped so significantly. The overlap

TABLE IV
5-FOLD CROSS-VALIDATION RESULTS FOR COMBINED DATASET

Model	Acc.	Prec.	Recall	F1	ROC-AUC
Convex Hull	0.4582	0.9490	0.0131	0.0258	0.5061
Convex Hull OutlierRM 1	0.6254	0.6101	0.8789	0.7202	0.5981
Convex Hull OutlierRM 2	0.6118	0.8185	0.3757	0.5150	0.6372
Convex Hull OutlierRM 3	0.4815	0.9041	0.0615	0.1151	0.5268
Logistic Regression	0.6383	0.6662	0.6828	0.6744	0.6335
kNN Scaled k=5	0.7949	0.8184	0.8048	0.8115	0.7939
Random Forest	0.7918	0.8165	0.8004	0.8084	0.7909
SVM Linear	0.6467	0.6871	0.6537	0.6700	0.6460
SVM RBF	0.7107	0.7650	0.6823	0.7213	0.7138

MPDD Benign vs Malicious Embeddings

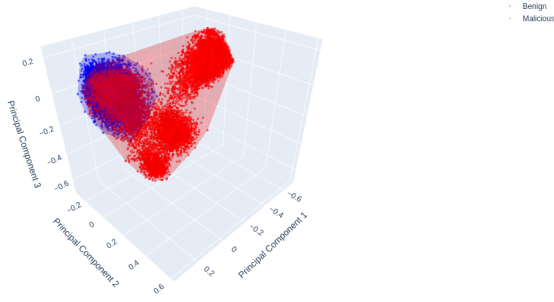


Fig. 1. MPDD PCA reduced embeddings for Benign and Malignant Prompts

Benign vs Malicious Convex Hulls By Dataset

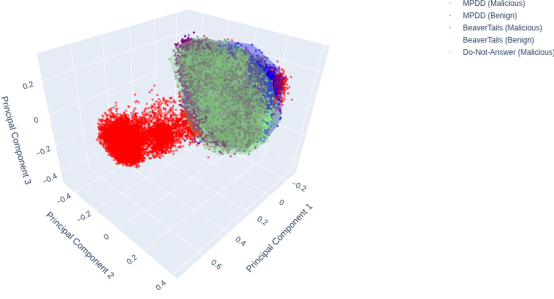


Fig. 2. All sets PCA reduced embedding for Benign and Malignant Prompts

of area between benign and malignant has increased as the convex hulls are almost immediately on top of one another as demonstrated in Figure 3.

Benign vs Malicious Convex Hulls By Dataset

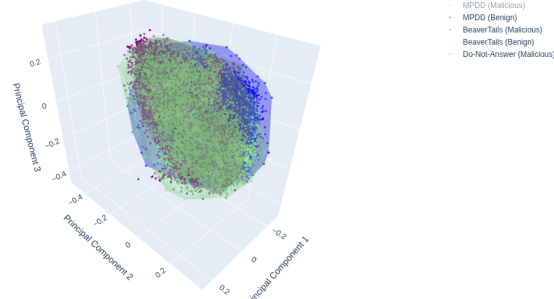


Fig. 3. All sets PCA reduced embedding for Benign and Malignant Prompts Overlapped

This overlap results in the malignant and benign embeddings existing in the same space leading to misclassifications. This also explains why the MPDD trained classifiers performed well when compared to the other experiments as the benign and malignant sets mostly occupied their own space. Reintroduction of higher dimensions may increase the separability of the two classes. Although the performance was close to traditional models with the MPDD, higher dimensional embeddings with less aggressive compression could increase the performance of convex hull classifiers. Optimization between

feature count and computational cost could provide developers with an improved method for malignant prompt detection.

VII. CONCLUSION

Convex hulls may provide a new method for classification of malignant prompt embeddings. Although outliers may artificially inflate the area of a computed convex hull, the outlier removal models improved performance. This however was not enough to surpass traditional models. The overlap of data points proved to be a difficult hurdle for convex hulls to overcome, but if further separation is possible performance may be improved. If data points from different classes cannot be separated then convex hulls may not be a viable solution as seen with the BeaverTails and Do-Not-Answer sets. Under specific conditions, where classes are clearly distinct from one another in latent space, convex hulls could be used as a defense guardrail against bad actors' input.

VIII. FUTURE WORK

Nemirko et al.'s work showed promising classification with approximations of convex hulls alongside distance calculation when points exist outside of class. Kudo et al, had similar results with their research using reflective convex hulls. Using a combination of these previous algorithms or methods could provide further improvement in malignant prompt classification. Alpha shapes are another potential classifier if the generality of convex hulls harms performance as they form a tighter geometric envelope of data points. A combination of convex hulls and traditional models could solve problems of overlap. This could be done by taking the difference between two convex hull classes and removing it. The two remaining convex hulls could act as two classifier hulls, while the overlap area could instead rely on traditional models for classification. It would also be interesting to measure the speed of both training and prediction for each of these models to see if convex hulls provide a benefit in speed.

REFERENCES

- [1] N. Maslej et al., "The AI Index 2025 Annual Report," AI Index Steering Committee, Institute for Human-Centered AI, Stanford Univ., Stanford, CA, USA, Apr. 2025. [Online]. Available: <https://hai.stanford.edu/ai-index/2025-ai-index-report/economy>
- [2] M. Faverio and O. Sidoti, "Teens, Social Media and AI Chatbots 2025," Pew Research Center, Washington, D.C., USA, Rep., Dec. 2025. [Online]. Available: <https://www.pewresearch.org/internet/2025/12/09/teens-social-media-and-ai-chatbots-2025/>
- [3] Federal Bureau of Investigation, "Criminals Use Generative Artificial Intelligence to Facilitate Financial Fraud," Internet Crime Complaint Center (IC3), Public Service Announcement I-120324-PSA, Dec. 3, 2024. [Online]. Available: <https://www.ic3.gov/PSA/2024/PSA241203>
- [4] W. Zhou et al., "EasyJailbreak: A Unified Framework for Jailbreaking Large Language Models," *arXiv preprint arXiv:2403.17336*, 2024.
- [5] N. Carlini and D. Wagner, "Adversarial Examples Are Not Easily Detected: Bypassing Ten Detection Methods," in *Proc. 10th ACM Workshop Artif. Intell. Secur. (AISec)*, Dallas, TX, USA, Nov. 2017, pp. 3–14.
- [6] C. Chuan, H. Ren, and P. Frossard, "AdvKNN: Adversarial Attacks on K-Nearest Neighbor Classifiers with Approximate Gradients," *arXiv preprint arXiv:1911.06591*, 2019.
- [7] E. W. Weisstein, "Convex Hull," *MathWorld—A Wolfram Web Resource*. [Online]. Available: <https://mathworld.wolfram.com/ConvexHull.html>. [Accessed: Dec. 14, 2025].

- [8] A. P. Nemirko and J. H. Dula, "Convex Hull Proximity Estimation for Machine Learning Problems," *Procedia Comput. Sci.*, vol. 186, pp. 187–194, 2021, doi: 10.1016/j.procs.2021.04.138.
- [9] M. Kudo, A. Nakamura, and I. Takigawa, "Classification by reflective convex hulls," in *Proc. 19th Int. Conf. Pattern Recognit. (ICPR)*, Tampa, FL, USA, Dec. 2008, pp. 1–4, doi: 10.1109/ICPR.2008.4761789.
- [10] A. P. Nemirko, "Convex Hull Proximity Estimation for Machine Learning Problems," *Pattern Recognit. Image Anal.*, vol. 32, no. 3, pp. 616–621, Sep. 2022, doi: 10.1134/S1054661822030282.
- [11] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Oct. 2011.
- [12] P. Virtanen *et al.*, "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nat. Methods*, vol. 17, pp. 261–272, 2020.
- [13] W. McKinney, "Data Structures for Statistical Computing in Python," in *Proc. 9th Python in Sci. Conf.*, Austin, TX, USA, 2010, pp. 56–61.
- [14] Plotly Technologies Inc., "Collaborative Data Science," Montreal, QC, Canada, 2015. [Online]. Available: <https://plot.ly>.
- [15] OpenAI, "New Embedding Models and API Updates," OpenAI Blog, Jan. 25, 2024. [Online]. Available: <https://openai.com/blog/new-embedding-models-and-api-updates>.
- [16] J. Ji *et al.*, "BeaverTails: Towards Improved Safety Alignment of LLM via a Human-Preference Dataset," in *Proc. 37th Conf. Neural Inf. Process. Syst. (NeurIPS)*, New Orleans, LA, USA, Dec. 2023.
- [17] Y. Wang *et al.*, "Do-Not-Answer: Evaluating Safeguards in LLMs," in *Findings of the Assoc. for Comput. Linguistics: EACL 2024*, St. Julian's, Malta, Mar. 2024, pp. 896–911.
- [18] M. A. Jebbar, "Malicious Prompt Detection Dataset (MPDD)," Kaggle.com, 2024. [Online]. Available: <https://www.kaggle.com/datasets/mohammedaminejebbar/malicious-prompt-detection-dataset-mpdd>. [Accessed: Dec. 14, 2025].